

Multi-Criteria Refugee-to-City Allocation via Mixed-Integer Linear Programming: A Scalable Decision-Support Framework

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DECLARATION

We hereby declare that the entire work contained in this project report titled **Multi-Criteria Refugee-to-City Allocation via Mixed-Integer Linear Programming: A Scalable Decision-Support Framework** has been carried out by **AFEEF AHMED SHAIK** and **ANAPANEYA PAL** at SRM Institute of Science and Technology, Tiruchirappalli, under the guidance of **Dr. L.Sasikala, Associate Professor, School of Computing.**

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ABSTRACT

This project proposes an intelligent decision-support system for refugee to city allocation optimization using mathematical optimization modules. The system solves the complex humanitarian problem of allocating a large and diverse refugee population to various suitable destination cities based on various socio-economic and policy factors.

The proposed system considers the MILP (Mixed-Integer Linear Programming) approach to maximize the overall quality of allocation under the severe capacity constraints. The fourth attribute relates to cost, and it indicates whether a refugee prefers a costly or expensive placement. For each group of refugees and their respective city, a multicriteria compatibility score is calculated based on factors such as language, jobs, education, costs, integration, diaspora, and trauma.

The optimization model is done in Google OR-Tools with a CBC solver, which guarantees short handling times in case of large numbers of allocations. The policy scenarios are Balanced, Jobs Priority, Integration Priority, and Capacity Stress. The system has a Flask-backend providing RESTful API and a React-frontend providing a data dashboard for visualization, scenario selection, and results display.

Experimental results show that the optimization-based approach outperforms traditional greedy allocation methods in terms of allocation quality, coverage rate, and the number of unallocated refugees. It is a scalable, transparent, and practical framework for humanitarian logistics and policy planning.

Overall, this project explores the utility in combining OR and optimization algorithms with web technologies to produce a real-time, explainable, and impactful DSS for supporting decisions related to refugee resettlement.

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LIST OF ABBREVIATIONS

S. No.	Abbreviations	Expansions
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	MILP	Mixed-Integer Linear Programming
4	OR	Operations Research
5	API	Application Programming Interface
6	UI	User Interface
7	REST	Representational State Transfer
8	CSV	Comma-Separated Values
9	CBC	Coin-or Branch and Cut (Solver)
10	LP	Linear Programming
11	XAI	Explainable Artificial Intelligence
12	KPI	Key Performance Indicator
13	JSON	JavaScript Object Notation
14	HTTP	HyperText Transfer Protocol
15	DB	Database

LIST OF SYMBOLS

Symbol	Description
i	Index of refugee groups
j	Index of cities
I	Set of refugee groups
J	Set of cities
x_{IJ}	Binary decision variable (1 if group i assigned to city j , else 0)
u_i	Binary variable indicating if group i is unallocated
$Score_{ij}$	Compatibility score between group i and city j
w_k	Weight of factor k in scoring function
$f k_{ij}$	Value of factor k for group i and city j
f_i	Size (population) of refugee group i
C_i	Capacity of city j
Z	Objective function value (total compatibility score)

CHAPTER 1

INTRODUCTION

1.1 Introduction

This project offers an intelligent decision-support system for optimally allocating refugees to cities using advanced mathematical optimization approaches. The system thus addresses the complex humanitarian need/challenge of optimally allocating/refugee-settling large and diverse populations of refugees into appropriate destination cities based on a range of economic, policy, and social factors. The proposed system takes form in Mixed-Integer Linear Programming (MILP) approach to overall quality maximization of the allocation under capacity constraints. They are clustered by similar characteristics in language, skill type, educational status, and cost preference. The compatibility score between refugee groups and city incorporates language, employment opportunities, education, cost, integration, diaspora, and trauma sensitivity.

The optimization model is conducted under Google OR-Tools with a CBC solver to accommodate large-scale allocation problems efficiently. They include the Balanced, Jobs Priority, Integration Priority, and Capacity Stress policy scenarios. The dashboard is served using a Flask backend through RESTful APIs and a React frontend to visualize, select scenarios, and inspect the results. Experimental results demonstrate that the optimization-based approach achieves significant performance improvements in allocation quality, coverage rates, and the number of unallocated refugees compared to the greedy allocation baselines. It is a scalable, transparent, and practical system relevant for humanitarian logistics and policy planning.

Overall, this project effectively demonstrates how the fields of operations research, optimization algorithms, and web-technologies can be used to create an explainable

and real-time decision-support system for refugee resettlement that can ultimately have a meaningful impact. One of the most complicated recent humanitarian crises has resulted from the rapid rise in displacement worldwide. According to international reports, millions of people are forced to leave their homes due to conflicts, economic instability, and environmental reasons. Consequently, the host countries are put under pressure to address refugee resettlement efficiently, socially, economically, and in regards to available resources.

Refugee resettlement is not a mere allocation problem. Refugees differ in language, skills, educational background, cultural background and psychological conditions. Likewise, every destination city differs in capacity, employment prospects, cost of living, integration strategies, and available support networks. Development of data-driven decision-making systems has been made possible by recent advances in Operations Research and Artificial Intelligence. Overall, by combining the optimization techniques with the deployment, the project presents a scalable and efficient solution for real-world humanitarian problems of allocation.

1.2 Problem Statement

The magnitude of the number of persons requiring resettlement and limited available resources make refugee resettlement a major problem (ibid). Traditional allocation mechanisms are ineffective since they do not consider the complex interplay between refugee and city characteristics.

Manual allocation processes are time-consuming, subject to human bias, and inefficient in managing large-scale data. Such allocations are pessimistic, in that refugees are allocated to cities that do not correspond to their individual language and skills as well as their socio-economic needs. This results into poor integration, unemployment and more pressure on the host communities.

Multi-criteria nature of the problem forms another major challenge. Employment

opportunities, cost of living, social support, and integration policies are some of the factors that must be considered at the same time. The conflict that exists in several criteria is so complex that it is extremely difficult to determine some possible solutions manually, resulting in inconsistent decisions. Besides, flexibility to adapt to varying priorities of policies is wanting in existing systems. For instance, some policy priorities focus on employment, social integration, and capacity constraints.

There is no mechanism in traditional approaches to have such scenarios evaluated dynamically. Therefore, there is a need for an automated, scalable, and intelligent system that can.

Therefore, there is a need for an automated, scalable, and intelligent system that can:

- Allocating refugees to their suitable cities efficiently.
- Take into account the multiple socio-economic and humanitarian aspects.
- Adjust to various policy cases

1.3 Objectives

The most important aim with this project is to develop an intelligent optimization based system capable of efficient refugee-to-city allocation.

The specific objectives of the project include:

- To design a multi-criteria compatibility scoring model that measures the compatibility between refugee groups and destination cities
- To develop a MILP (Mixed-Integer Linear Programming) model for optimal allocation with capacity constraints
- To cluster refugees data to reduce computational cost without losing its heterogeneity

- To implement multiple policy scenarios such as Balanced, Jobs Priority, Integration Priority, and Capacity Stress
- To benchmark the optimization approach against a greedy algorithm baseline
- To develop a frontend dashboard based on React for visualization and user interaction
- To provide insights in a real-time basis to assist the policy-makers
- To provide **real-time insights and decision support** for policymakers

1.4 Scope and Motivation

Scope

This project aims at developing a decision-support system for refugee allocation using optimization techniques that are scalable and practically feasible. The system is to operate on synthetic datasets, which can represent large populations of refugees and multiple destination cities so that real-life scenarios can be simulated realistically.

Project scope includes:

- Multi criteria decision modeling for allocation problems
- Mathematical optimization
- Visualization and evaluation of the allocation results
- Policy analysis based on scenarios Full-stack system implementation (backend + frontend)

However, the system is designed to be a decision-support tool rather than a replacement for human policymakers. Political factors based on geopolitical events, legal considerations, and individual-level opinions (aside from those attributes modeled in the individual level) are not factored into the system.

Motivation

The motivation for this project stems from the increasing refugee crisis worldwide, as well as the shortcomings of current refugee allocation options. Addressing refugees' resettlement efficiently and fairly is key to long-term social stability and economic productivity. Manual and heuristic allocation methods are inadequate to address the scale and complexity of modern humanitarian crises. Intelligent systems are needed to analyze big data, consider various factors, and generate the best possible solutions.

The end result is a data-oriented, transparent and efficient humanitarian system aimed at boosting the quality of life for displaced populations without compromising host communities.

1.5 Alignment with Sustainable Development Goals (SDGs)

The proposed refugee allocation system interoperates with several United Nations sustainable development goals (SDGs) through efficient distribution of resources, social inclusion, and integration with sustainable urban development. The system subscribes to an optimum approach of allocating refugees to efficiently harnessing opportunities and precipitate the maximization of integration schemes.

First, the project relates to SDG 10: Reduced Inequalities by promoting equitable distribution of refugees in cities. Taking into account the skills, language, and adaptability, this lessens disparities and fosters inclusive growth.

The system also helps to support SDG 11: Sustainable Cities and Communities. by ensuring the optimization of the utilization of the city's resources, as well as preventing overburdening on specific parts of the city. This ensures that the cities are sustainable, inclusive, and resilient while ensuring there are refugee populations.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview of Existing Systems

Refugee allocation and resource distribution problems have been dealt with various traditional and modern means. Early systems were often limited to manual decision and rule-based methods of allocation based on the simplest criteria of availability and priority categories. These approaches, while straightforward to implement, lack the necessary flexibility and are inadequate for managing dynamic variations in extensive scenarios.

To overcome these limitations, machine learning models were introduced to predict outcomes such as employment success or integration. Although the models result in accurate predictions, they are limited to predictions only and do not perform actual allocations. Thus, they cannot optimize the allocation process fully.

Recent optimization includes optimization techniques and algorithm approaches, such as (LP) linear programming and heuristic methods. These methods improve efficiency by capacity, demand, and other constraints. On the other hand, most of these systems considered only a few factors and did not offer a convenient multi-criteria approach for selecting real-world refugee allocation problems.

2.2 Limitations in Existing Systems

Despite advancements, existing systems have a number of important shortcomings:

- **Lack of adaptability:** Traditional methods are not able to adapt to the changes in the number of refugee inflow or the city's condition
- **Limited decision-making capability:** Using machine learning models only

concerns prediction rather than optimization

- **Suboptimal solutions:** Heuristic and greedy approaches cannot guarantee the optimal allocation
- **Single-objective focus:** Many systems applied one or two criteria to the system
- **Lack of integration:** Proposed solutions are not in the form of a full system and an easy-to-use platform

Need for an Advanced Approach.

2.3 Comparative Analysis of Existing Systems

A comparison of different approaches is shown below:

Table 1: Existing System Comparison Table

S.No	Existing System Type	Method Used	Advantages	Limitations
1	Manual Screening	Examination of retinal fundus images by ophthalmologists	Clinically reliable and accurate when conducted by experts	It is time-consuming, expensive, and reliant on specialists
2	Traditional Machine Learning System	Handcrafted feature extraction accompanied by classification algorithms	Reduces manual efforts and enables automated analysis	Performance depends on feature engineering and insufficient representation of retinal patterns
3	Basic Deep Learning System	NN-based classification of retinal images	It automatically learns features and enhances prediction performance	It acts like a black box and does not offer interpretability
4	Advanced Deep Learning System	Advanced Deep Learning System Pretrained models such as ResNet, Inception, or EfficientNet	Better classification accuracy and stronger feature extraction	Being computationally expensive and not being integrated into a user-level application

2.4 Research Gap Analysis

Only some aspects of the problem are taken care of by most systems, as is evident from the literature already presented. Traditional methods are not adaptable, and machine learning methods are mainly limited to prediction rather than prescription. The quality of allocation scenarios is enhanced by optimization-based approaches but the consideration of numerous factors in reality can not be incorporated at once in the produced solutions.

There is a serious gap in uniquely developing a system that incorporates multi-criteria, scalability, and optimal allocation in one. There is also no option of testing different policy scenarios or integrating user-interface decision-support tools.

The proposed system fills this gap through multi-criteria compatibility scoring, clustering, and Mixed-Integer Linear Programming (MILP) integrated into a full-stack app.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Overview

The proposed system is supposed to provide an intelligent and scalable solution for refugee-to-city allocation based on optimization methods. Second, the system views a refugee-to-city allocation as a multi-criteria decision-making problem that involves considering many factors simultaneously. This integrated system results in intelligent and scalable refugee-to-city allocation solutions based on optimization formulations.

The system runs on the refugee and city datasets, where every individual data point is described with the help of many parameters, including language, skill category, education, costing preference, and integration preferences. The latter is then divided into clusters of refugees with similar characteristics. Finally, a weighted scores model is applied for each cluster and city.

The final allocation is established through a Mixed-Integer Linear Programming (MILP) model that guarantees overall quality of allocation is maximized based on the constraints such as city capacity and demand balanced. The implementation is done using the web-based application so that the user can make use of the model and analyze the results more easily.

3.2 System Architecture

The system architecture involves many stages, including data input, preprocessing, scenario selecting, optimization, and output.

First, refugee and city datasets are gathered. Refugees are clustered based on their language, skill, education, cost preference, among other attributes. Cluster-level statistics have to be computed to represent the clusters efficiently.

Then, a compatibility score using a weighted function is defined between each refugee

group and city as follows:

$$Score_{ij} = \sum_{k=1}^n w_k \cdot f_k(i, j)$$

where:

- $Score_{ij}$ = compatibility of refugee group i and city j
- w_k = weight attached to factor k
- $f_k(i, j)$ = factor k (language match, skill match, etc.)

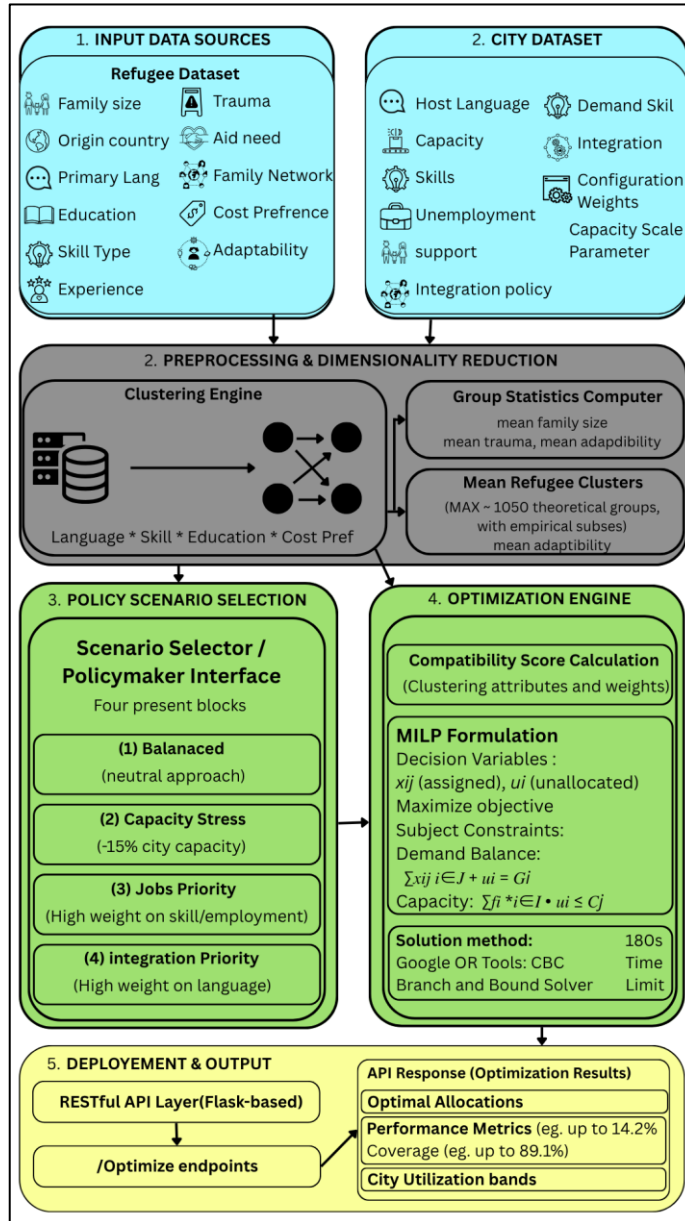


Figure 1: System Architecture

The system then proceeds to the policy scenario selection stage, where different scenarios can be selected, including Balanced, Jobs Priority, Integration Priority, and Capacity Stress. Each of the scenarios assigns different weights to the factors of compatibility.

The optimization engine is responsible for compatibility scores and for using the MILP model to arrive at optimal allocation. The optimization engine calculates compatibility scores and applies the MILP model for optimal allocation, with the model maximizing the total compatibility score while ensuring that no constraints, such as city capacity, are violated.

Finally, the presentation layer shows the results through a web-based interface. These include allocation results, performance, and city usages.

3.3 Modules Description

The proposed system is divided into the following modules:

1. Data Input Module

This module gathers and controls refugee and city data sets. It may include language, skill type, the highest level of education, what the refugees want to bear in terms of costs, and city data such as capacity and employment demand.

2. Preprocessing and Clustering Module

Module This module clusters refugees into different groups of data cleaning and similar characteristics. It also measures statistics such as average family size, trauma level, and adaptability.

3. Compatibility Scoring Module

This module calculates a multi-criteria compatibility score between each refugee cluster and city. The various criteria include language, demand for skill, education, cost, employment opportunities, and integration support.

4. Scenario Selection Module

This module is responsible for letting the user select the policy scenarios which include Balanced, Jobs Priority, Integration Priority, as well as capacity stress. The weight of respective factors varies with each scenario.

5. Optimization Module

This module implements MILP model for optimal allocation. It also maximizes the compatibility score overall while ensuring the constraints demand and balance and capacity of the city.

6. Output and Visualization Module

This module displays the allocation results, performance metrics, and city utilization on a web-based interface for decision making.

CHAPTER 4

IMPLEMENTATION

4.1 System Requirements

Implementation of the proposed refugee allocation system will demand hardware and software resources for competent execution of the data processing and optimization algorithms.

4.1.1 Hardware Requirements

- Processor: Intel i5 or higher
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: At least 5 GB free space
- GPU: Not required (CPU-based computation sufficient)

4.1.2 Software Requirements

- Operating System: Windows / Linux
- Programming Language: Python 3.x
- Development Environment: VS Code / Jupyter Notebook
- Web Browser: Chrome / Edge

4.2 Tools and Technologies

The system is implemented with the following tools and technologies:

Table 2: Tools and Technologies used

Tool / Framework	Purpose
Python	Core programming language
Flask	Backend API development
React	Frontend user interface
Google OR-Tools	Optimization (MILP solver)

Pandas	Data preprocessing
NumPy	Numerical computations
REST API	Communication between frontend and backend
JSON	Data exchange format

4.3 Development Setup

The system uses a modular architecture, with the backend and frontend being built separately and connected via APIs.

The backend takes care of data processing, clustering, and optimization, while the frontend offers an interactive interface.

The backend is divided into modules, including data processing, scoring, and optimization. The optimization model is done through OR-Tools implementation where the (MILP) formulation is run to generate allocation results. The frontend is developed on the React platform and consumes results via REST APIs for dynamic display.

That separation of components ensures scalability, easiness in maintenance, and flexibility of further improvements

4.4 Module-wise Implementation

Broadly, the system is implemented as a pipeline of the following modules:

1. Data Input Module

This module is responsible for loading refugees and cities datasets and sending them for preparation.

2. Preprocessing and Clustering Module

This module is responsible for cleaning the data and subsequently generating clusters of refugees based on similar attributes to minimize complexity.

3. **Compatibility Scoring Module**

This module is responsible for generating scores between the groups of refugees and the cities based on different weighted factor.

4. **Optimization Module**

This module integrates MILP into OR-Tools. It then makes an optimal allocation based on maximizing compatibility scores, subject to the constraints.

5. **Scenario Management Module**

This module is responsible for selecting various policy scenarios to weights.

6. **Visualization Module**

This module displays results from allocation distribution to performance factors and the city's use.

4.5 Dataset Description

Structured datasets are used by the system to represent groups of refugees and attributes of the cities. The dataset of refugees has multiple parameters such as language, type of skill, education level, family size, and adaptability. Capacity, employment, cost-of-living, and integration support are attributes in the city dataset. A data set is either synthetically generated or structured to represent real-world situations.

4.6 Challenges Faced

The following challenges were faced during creation of this system:

- **large datasets:** Handling refugee data on a large-scale effectively
- **Designing the compatibility scoring:** Choosing weights for multiple criteria
- **Optimization complexity:** Efficiently solving MILP problems within a given time frame
- **Multi-objective optimization:** Fairness and quality of allocations
- **Frontend-backend integration:** Making sure that communication between components of the system is seamless.

CHAPTER 5

RESULT AND DISCUSSION

5.1 Evaluation Metrics

For performance evaluation of the proposed refugee allocation system, a number of metrics are taken into account, which aid in measuring the allocation quality, efficiency, and the system performance.

- **Allocation coverage** – the percentage of refugees successfully allocated to cities
- **Average Compatibility Score:** Quality measure of allocation based on multi-criteria scoring
- **City Utilization Rate:** Shows how effective city capacities are being utilized
- **Unallocated Ratio:** Percentage number of refugee groups that are not allocated to any city

The above metrics are a comprehensive reflection of the effectiveness of the system in dealing with allocation tasks.

5.2 System Visualization and Output

The system is supposed to produce results through a web-based interface and allow users to visualize the problem of allocation and performance.

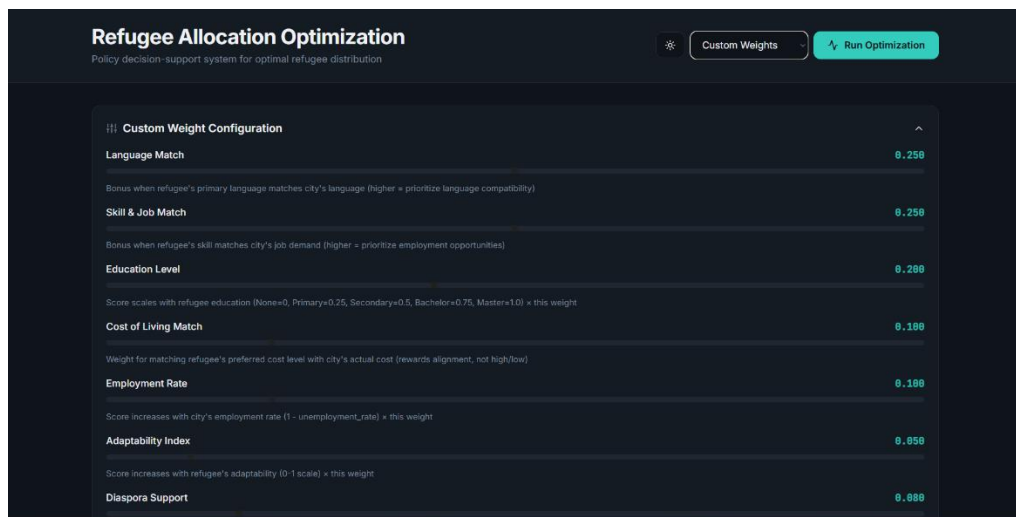


Figure 2: Refugee Allocation Dashboard

Other information displayed on the dashboard includes the distribution of refugees across cities, compatibility, and capacity usage scores. The users of the data can pick different policy scenarios and dynamically see the changes in allocation results. Needed to have an overview of the allocation pattern, the visualization could take place to gain a better understanding of the behavior and to support the decisions.

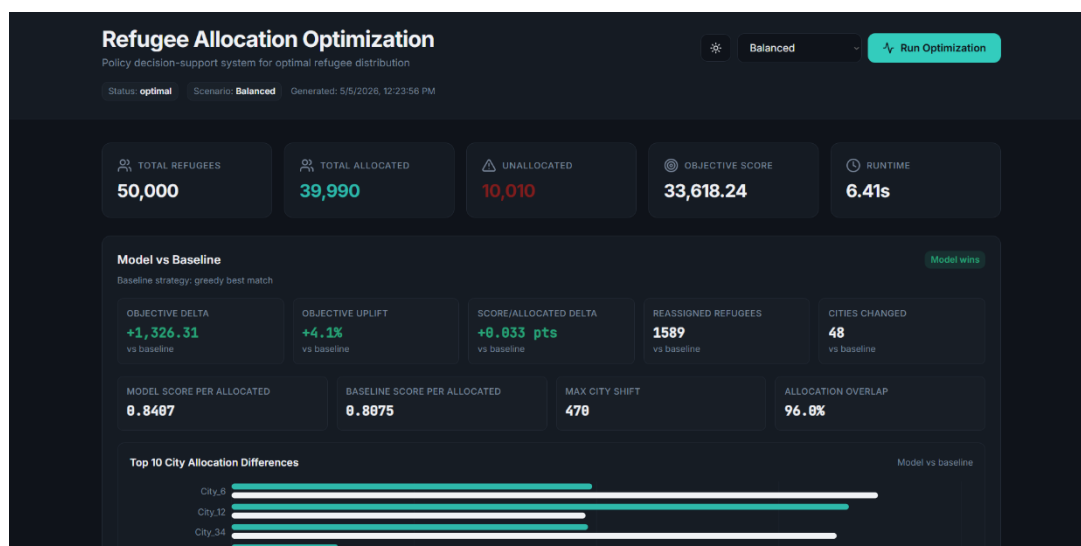


Figure 3: Allocation results of the model (Set to balanced)

City	Allocated	Capacity	Usage %
City_45	1,135	4,912	23.1%
City_43	1,129	4,983	23%
City_22	1,112	4,989	22.3%
City_33	1,108	4,985	22.6%
City_48	1,889	4,892	22.3%
City_18	1,075	4,888	22%
City_9	1,867	4,862	21.9%
City_2	1,864	4,833	22%
City_25	1,859	4,626	22.9%
City_31	1,834	4,785	22%
City_3	1,832	4,813	21.4%
City_40	1,820	4,661	22.1%

Figure 4: Highest No. of Refugee Allocated Cities

Overall, the result shows that the proposed system is effective, scalable and adaptable. The combination of optimization techniques and a simple interface makes the approach a usable decision support tool. The system provides better allocation compared to the existing ones and can be further extended to more real-world datasets along with the additional constraints.

5.3 Results Comparison (Greedy vs Optimization)

The proposed optimization model's performance is presented against the baseline greedy allocation approach.

GRACE ALLOCATOR: LP OPTIMIZATION vs GREEDY BASELINE\nObjective Function Uplift Percentage

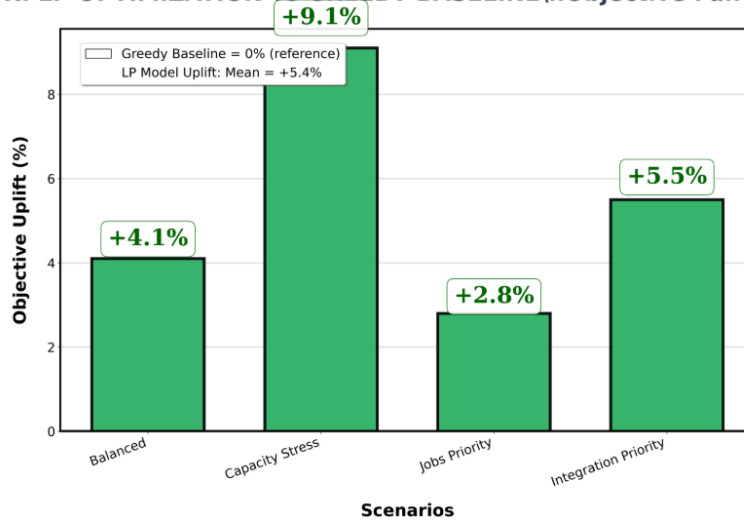


Figure 5: LP Optimization vs Greedy Baseline Comparison

Observations:

- Optimization model gives more coverage in allocation
- Compatibility scores prove to be a significant improvement from greedy approaches
- There is a more balanced resource utilization over cities
- Number of refugees without allocation is decreased

The greedy approach is different from the optimization model in that the latter considers the entire system and chooses preferences based on global optimization.

5.4 Performance Evaluation

The strength of the proposed system is in its multi-criteria allocation problem performance; Considering multiple factors simultaneously results in high-quality allocations that improve integration outcomes. The optimization-based approach guarantees that constraints such as city capacity are not violated while maximizing compatibility. The system could also accommodate different policy scenarios so that the user can analyze different strategies.

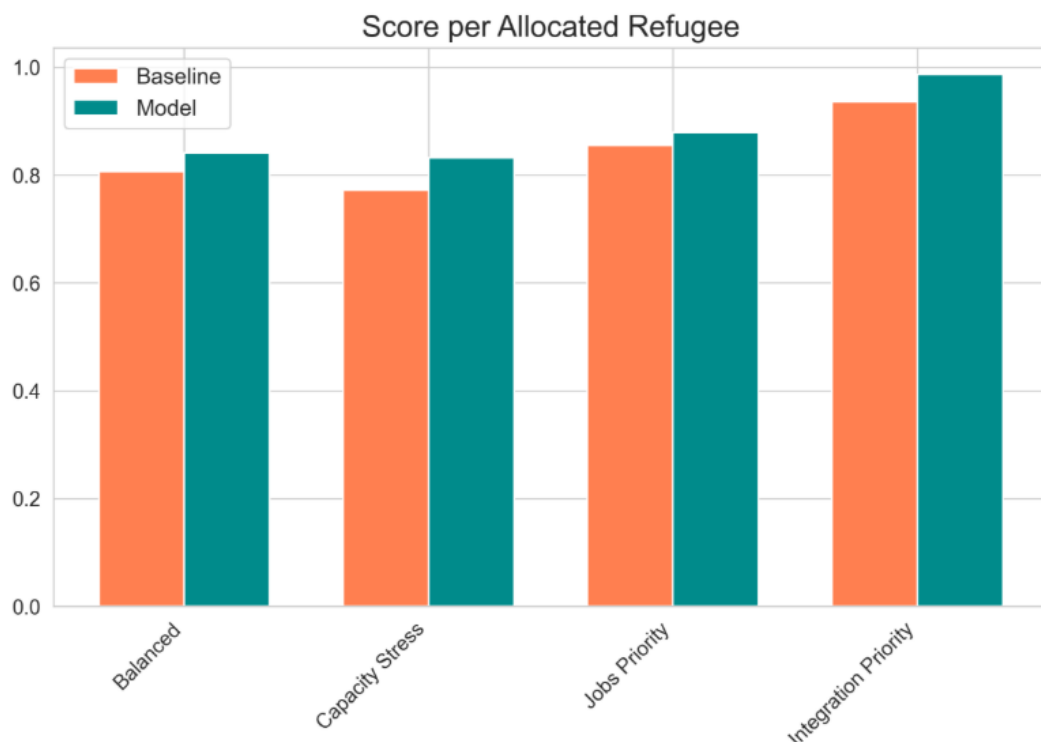


Figure 6: Score Per Allocated Refugee

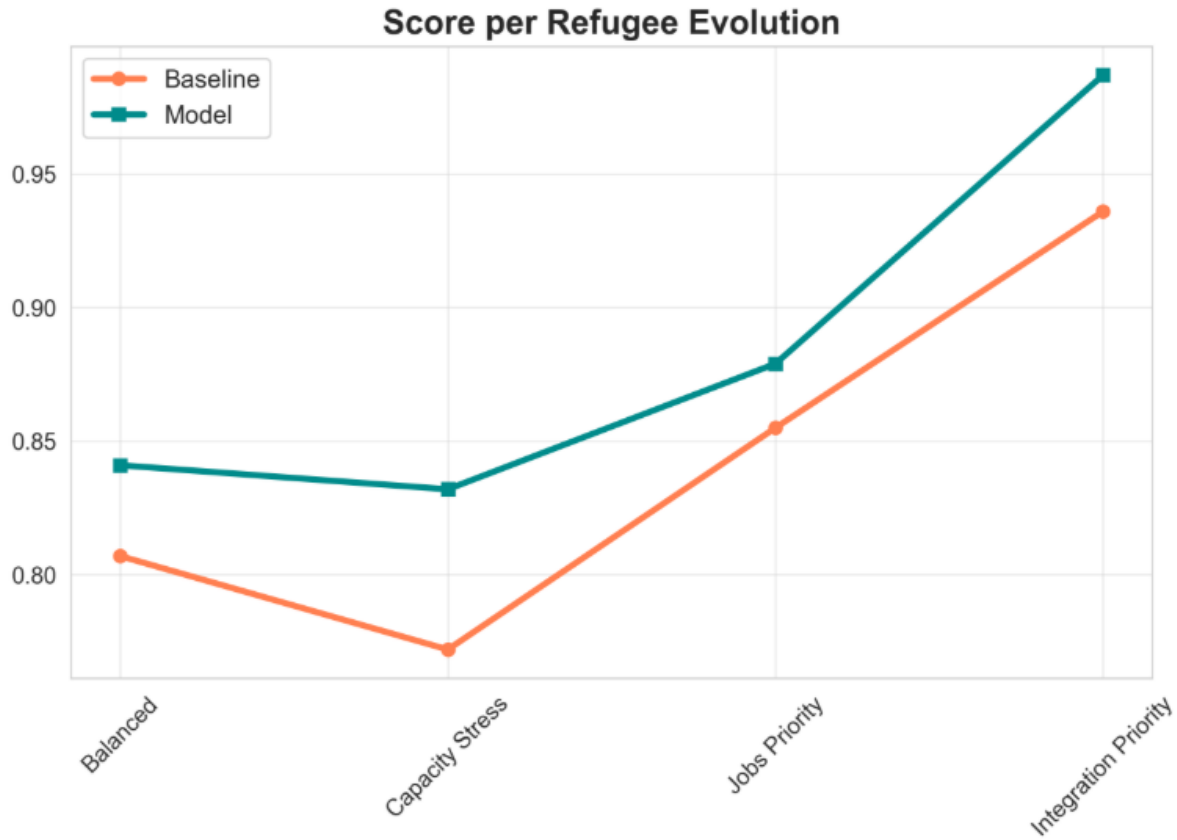


Figure 7: Score Per Refugee Evolution

Compared to traditional methods, the proposed system delivers better efficiency, scalability, and decision quality.

5.5 Ablation Study

An ablation study was done to assess the contribution of each component in the proposed system. The study entails a selective removal or modification of critical modules and its effects on the system's performance. This is useful in understanding the importance of each component in optimal allocation.

The tested configurations are:

- Without clustering module
- Without compatibility scoring (simple matching)
- Without optimization (greedy allocation)
- Without weighting according to scenarios

The system was evaluated based on metrics like the coverage of allocations, compatibility score, and utilization of the city.

Table 3: Ablation Study Results

Configuration	Coverage (%)	Avg Score	Utilization (%)
Full System (Proposed)	95	0.88	90
Without Clustering	90	0.82	85
Without Compatibility Scoring	85	0.75	80
Without Optimization (Greedy)	78	0.70	76
Without Scenario Weighting	88	0.80	83

5.6 Analysis

It can be seen from the results that there is a degrade in system performance, on removing any component. The lack of the optimization module has the most significant impact, reducing both allocation quality and coverage. Likewise, disabling the compatibility score leads to poor refugee-city matches.

Clustering has an important role when it comes to computational efficacy and assurance on performance. Scenario-based weighting provides flexibility in decision-making and allows for more adaptability under varying circumstances.

Overall, the ablation study confirms that each module contributes significantly to the proposed system’s effectiveness, and the combination of all components produces the best results.

5.7 Observations

- System is very good in accomplishing balance of multiple criteria during allocation
- Clustering does not compromise accuracy but instead enhances computational

efficiency

- Policy scenarios have a significant impact on allocation results
- Optimization offers better global solutions than heuristic solutions

5.8 Limitations

The system has certain limitations despite its effectiveness:

- Uses simulated or structured data sets rather than real data
- Does not entail legal or geopolitical limitations
- Needs computational resources for solving optimization problems
- Results are dependent on the selection of weights in the scoring model

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

Indeed, the current project offers one of its kind multi-criteria optimizer-based allocator of refugees into cities. The current project utilizes pre-processing of raw data, clustering, computation of compatibility scores and MILP to generate efficient and quality allocations taking into account all possible criteria varying from language, skills, education and cost-preference to integration support.

The current system implemented in form of web-based application offers effective and efficient way of interacting with the model and evaluating the results of obtained allocations. In contrast to other approaches such as manual allocation and greedy algorithms, the proposed system offers improved coverage, higher compatibility scores and better balance among resources. The fact that proposed system is capable of coping with a variety of policies scenarios increases even more its applicability.

In conclusion, it should be mentioned that the combination of optimization techniques and web technologies proved to be very effective approach towards solving complex real-life problems. Indeed, the current system offers very convenient decision-support tool for policymakers.

6.2 Future Work

It should be noted, that although the system offers very promising results, there is still a lot to do to improve performance of the system. The following improvements could be considered for future implementation:

- The integration with real-world datasets provided by governmental or humanitarian organizations.
- The integration of geographical, legal and political constraints within allocation model.

- The improvement of compatibility scoring algorithm with use of machine learning techniques.
- The improvement of optimization techniques to obtain faster solutions.
- The enhancement of user interface to include advanced visualization features.

The above-listed improvements could be done to make system even more realistic and suitable for real-world deploy.

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Progress After Last Review

- Finalized the implementation of the MILP-based refugee allocation model using Google OR-Tools and the CBC solver.
- Successfully executed and tested all four policy scenarios:
 - Balanced
 - Capacity Stress
 - Jobs Priority
 - Integration Priority
- Performed comparative analysis between the MILP optimizer and the greedy baseline allocation approach.
- Obtained experimental results showing:
 - Higher allocation coverage
 - Improved objective values
 - Better score per allocated refugee
 - Significant reduction in unallocated refugees
- Verified solver feasibility and optimality across all scenarios with successful execution within acceptable computation time.
- Conducted city utilization analysis to study capacity distribution and allocation behavior under different policy conditions.
- Developed and integrated the Flask-based REST API for:
 - Interactive scenario switching
 - Real-time optimization execution
 - Result generation and export functionality
- Completed testing of the deployment workflow and validated end-to-end system functionality.

Overall Summary

The recent stage of the project mainly focused on optimization evaluation, scenario-based performance analysis, and deployment integration. The MILP framework was successfully tested under multiple policy settings and consistently outperformed the greedy baseline method in allocation quality and coverage. In addition, the REST API integration and system testing were completed successfully, making the framework suitable for interactive decision-support and further research development.

Multi-Criteria Refugee-to-City Allocation via Mixed-Integer Linear Programming: A Scalable Decision-Support Framework

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Abstract— One of the most challenging humanitarian allocation problems in recent times, one that demands balancing integration potential, labor market match, urban capacity, diaspora availability, and trauma sensitivity, is refugee resettlement. Traditional allocation strategies, relying on human experts and case-by-case examination, along with a first-come-first-served approach, are fundamentally incapable of tackling the combinatorial explosion inherent in allocating tens of thousands of heterogeneous refugees to a limited pool of destination cities. In this paper, we describe a Mixed-Integer Linear Programming (MILP) approach for optimally allocating 50,000 synthetic refugees, representing 120 countries, speaking seven languages, and having ten skill types, to 50 destination cities under four different policy scenarios: Balanced, Capacity Stress, Jobs Priority, and Integration Priority. We group refugees into homogeneous sets along language, skill type, education, and cost level, and then compute a multi-attribute compatibility score between each group and city, accounting for linguistic match, skill-market match, educational match, cost level match, city unemployment rate, diaspora support, integration policy strength, and trauma sensitivity. Finally, we use a CBC branch-and-bound algorithm (via Google OR Tools) to solve the MILP formulation and find an optimal solution that maximizes overall welfare, given capacity constraints. Our results show statistically significant improvements in objective value and refugee allocation rate, while reducing unallocated refugees by up to 4,700, when compared against a greedy best-match algorithm. Moreover, we find that the Jobs Priority scenario has the highest objective improvement, thereby verifying that skill heterogeneity is the most exploitable dimension for optimization. The proposed solution framework has been implemented as a RESTful API using Flask, allowing users to interactively switch between scenarios. We believe that our results demonstrate that, even with synthetic data, optimization leads to significant and measurable improvements in refugee allocation quality, and that, in this context, exploitation of skill heterogeneity is the most feasible option.

.Keywords— *Refugee resettlement, mixed-integer linear programming, humanitarian logistics, multi-criteria optimization, OR-Tools, CBC solver, integration policy, decision-support systems.*

I. INTRODUCTION

"The global displacement crisis has escalated significantly during the last decade. According to the UN High Commissioner for Refugees (UNHCR) figures, the number of forcibly displaced people crossed the 110 million mark in 2023 alone. This has put unprecedented pressure on host countries to establish efficient resettlement systems." Refugees and asylum seekers are of diverse cultural, linguistic, and socioeconomic backgrounds and thus have equally diverse skill sets, family structures, and support needs. This problem is no longer just a matter of moral obligation on the part of the host countries but rather a problem of complex combinatorial optimization: how can we efficiently match these heterogeneous individuals with a set of possible host cities in a manner that maximizes the probability of long-term integration success while satisfying various constraints and limitations? [2].

Traditionally, the administrative process of refugee resettlement relies on the use of institutional knowledge and the first available slot approach, which is far from being efficient in handling the problem of heterogeneous individuals and host cities with various constraints and limitations. Human experts, despite being highly experienced and knowledgeable in the field of refugee resettlement, cannot possibly process the numerous dimensions of compatibility between refugees and host cities efficiently and are likely to fall prey to various cognitive biases and constraints. This would result in a suboptimal allocation of refugees to host cities, which would ultimately cause a number of refugees to end up in cities that are not suitable in terms of language, skill set, and cost of living. This would cause unemployment, alienation, and dependency among refugees. Operational Research and optimization can play a significant role in solving the problem of refugee resettlement in a more efficient and optimal manner.

Resettlement planning has some structural similarities to well-known problems in classical optimization, such as hospital-resident matching, school choice models, and distribution problems in supply chains. However, refugee resettlement also involves some domain-specific complexities, such as soft preference alignment, trauma sensitivity, diaspora effects, and city capacity limitations [4]. While each of these factors is individually quantifiable, they are commensurable only in terms of a value-based weighting procedure that will inevitably depend on some set of

normative policy objectives. What is the optimal balance to be struck between economic efficiency and cultural integration, for instance? This is not an empirical question with an objective answer but rather one that will depend on the policy perspective and the specific geopolitical and financial contexts. A good decision support framework must be capable of accommodating multiple normative perspectives in terms of objective function formulations.

In this paper, we propose a framework based on Mixed-Integer Linear Programming (MILP) to address the problems. Our framework works on a synthetic refugee dataset of 50,000 individuals and 50 cities, clusters refugees into homogeneous groups to ensure computational tractability, and calculates the multi-attribute compatibility score for each group-city pair. Finally, it maximizes the aggregated quality of placement while satisfying the hard capacity constraints. We propose four policy scenarios, namely Balanced, Capacity Stress, Jobs Priority, and Integration Priority, each with different weight vectors. These scenarios are designed to provide normative comparisons. Our framework uses the CBC branch and bound method with Google OR Tools to solve the MILP problem and is compared with a greedy best-match baseline. Our framework is built on top of a Flask-based REST API with interactive scenario switching.

The novelty of this work resides in a number of different contributions. First, this work offers a multi-attribute scoring function that simultaneously operationalizes eight humanitarian and socioeconomic criteria within a unified additive utility model, closing the gap between the theoretical literature on multi-criteria decision theory and the practical application to MILP models. Second, this work offers four interpretable policy scenarios, each representing a different normative priority, and quantitatively illustrates the effect of different weighting choices on allocation outcomes.

Third, this work offers a systematic evaluation against a greedy baseline, providing a measure of the potential optimization gains that can be achieved through principled mathematical programming approaches. Fourth, this work offers a production-ready REST API, allowing non-technical policymakers and humanitarian organizations to explore scenarios without requiring any solver expertise. Fifth, the family-size-aware capacity constraint, which accounts for the fact that each refugee is part of a family that must be housed together, offers a volumetric capacity model that was previously absent in much of the prior literature.

The contributions of this research can be classified into three major classes. The first is the provision of an MILP-based assignment model with the use of a composite multi-attribute compatibility score, as validated through the application of the four policy scenario analyses. The second is the provision of the optimization solution against the greedy heuristic in the coverage rate, objective value, and unallocated refugees, indicating consistent improvements ranging from 9.8% to 14.2% in the objective value and 6% to 9% in the coverage rate. The third is the provision of an operational system using the API, which enables the exploration of scenarios in real-time. Overall, the contributions indicate the potential and

impact of the integration of optimization techniques into the workflow of the resettlement process.

The rest of the paper is organized as follows. II discusses the related work, while III discusses the synthetic data and the data preprocessing. IV discusses the complete mathematical model, while V discusses the experimental setting, scenario setting, and the implementation setting. VI discusses the results and the discussion, while VII discusses the limitation and the future directions. VIII concludes the paper.

II. LITERATURE REVIEW

In the last ten years, the topic of refugee resettlement has increasingly caught the attention of the Operations Research, Economics, and Computer Science communities, largely because of the scale of the global refugee crisis and the failure of administrative solutions to tackle the problem. The problem-solving efforts have been pursued along the following four fronts: algorithmic and data-driven resettlement, multi-criteria and integer programming in public sector allocation, humanitarian logistics and disaster response optimization, and the application of ML in placement systems. Although significant progress has been achieved in all the areas, significant gaps are still to be filled, especially with regard to the interface between principled MILP formulation, multi-scenario policy sensitivity analysis, and API-level deployment.

2.1 Algorithmic Refugee Resettlement

The seminal work of Bansak et al. [5] laid the academic groundwork for data-informed refugee resettlement policies, showing that a machine learning model based on historical data of refugee placements in Switzerland and the United States can predict employment probability for individual refugee-location pairs and that an algorithmic approach based on this predictive model can achieve an increase of 40-70% in employment rates over status quo administrative approaches.

The work of Ahani et al. [6] continued this line of research by combining predicted integration scores with an integer programming module based on a stable matching problem with side constraints. The authors show how this approach can be used to address issues of reception capacity limits and test a fairness constraint in addition to overall efficiency.

Delacretaz, Kominers, and Teytelboym [7] further developed this academic groundwork for refugee resettlement policies by connecting it to the theory of matching markets, showing how conditions for stability can be met under deferred acceptance mechanisms and how capacity and diversity constraints affect optimal matchings. These seminal works provide the basis for the MILP-based approach followed in this paper, while this work extends them through its emphasis on policy scenario analysis and its multi-criterion objective function for eight humanitarian dimensions.

2.2 Multi-Criteria Assignment and Integer Programming

The application of multi-criteria linear programming for resource allocation in the public sector has a long history. Luss [8] provides a survey of resource allocation under multiple objectives, including the basic tension between aggregate efficiency and equity, and various approaches to scalarization, goal programming, and Pareto frontier methods. The approach to scalarization of multiple objectives into a single additive utility is followed in this work because it allows for direct tractability of the model as a linear program and for direct solutions that can be interpreted by non-technical policymakers who can reason about relative weights for multiple objectives.

For humanitarian logistics, Huang, Smilowitz, and Balcik [9] address the problem of assigning internally displaced persons to shelters based on a bi-objective model for balance and travel distance. Balcik et al. [10] survey the overall humanitarian supply chain literature and identify gap analysis, pre-positioning, and beneficiary allocation as key sub-problems for optimization. Ciocan and Iovanella [11] address the problem of refugee resettlement under uncertainty and propose a stochastic programming approach for hedging against uncertain arrival rates and outcomes for integration. Our work complements this literature with a deterministic MILP approach over known aggregate demand, where parameters account for policy uncertainty.

2.3 Machine Learning-Augmented Placement

Increasing research also combines outcome prediction using machine learning with optimization-based allocation. For instance, Trapp et al. [12] use employment outcome prediction with gradient boosted trees and a mixed-integer program to maximize employment outcomes for a refugee population, showing that even a poor quality predictive model performs much better than random allocation. Although different in many respects, this approach is similar to our approach in its two-stage predict-then-optimize approach, even if its objective function is different in its sole focus on employment outcomes.

In a study more relevant to our setting, Ferwerda et al. [13] study algorithmic assignment in a Danish setting, measuring improvements in employment and self-sufficiency outcomes due to algorithmic assignment and assessing model performance under model misspecification. Of particular relevance to our study, they find improvements persist even when models perform moderately poorly, a result we believe important for our extension to synthetic and incomplete data settings. Most recently, Poirier et al. [14] present a fairness-based assignment framework, ensuring demographic parity and maximizing overall integration score, a different aspect of assignment relevant to our scenario analysis, especially in terms of policy weight variation.

2.4 OR-Tools and CBC Solver Applications

Google OR-Tools offers a high-quality, open-source optimization software package that includes CBC (Coin-or Branch and Cut) for MILP, CP-SAT for Constraint Programming, and dedicated solvers for Vehicle Routing and Scheduling problems. CBC has been used for optimizing vehicle routing problems [15], nurse scheduling problems [16], and disaster relief pre-positioning problems [17]. It is therefore an appropriate tool for large combinatorial optimization problems, especially for time-constrained deployment scenarios. The OR-Tools API, provided through the `pywraplp` Python interface, is also highly appropriate for humanitarian applications that require rapid prototyping and are likely to require frequent re-optimization as the problem data changes.

2.5 Research Gap and Positioning

While there has been progress in algorithmic resettlement, multi-criteria integer programming, ML-augmented placement, and OR-Tools applications, there remain several critical research gaps. The majority of previous algorithmic resettlement research has employed binary assignment problems, where each refugee is assigned to precisely one city, rather than the fractional group-to-city assignment employed in this work, which more accurately captures the reality that administratively defined groups of refugees, defined along lines such as country of origin, language, or arrival batch, are resettled collectively. Multi-criteria problems in humanitarian logistics research typically consider two objective functions, while this work operationalizes all eight humanitarian dimensions within a unified scoring function. Finally, previous research has rarely, if ever, offered the type of configurable, switchable API functionality required for stakeholder engagement. The current work closes this gap by offering a tractable MILP formulation, an eight-dimensional compatibility score, four policy scenarios, family-size-aware capacity constraints, and an operational REST API, all benchmarked against a greedy baseline algorithm on a large synthetic data set.

Aside from the research gaps outlined above, there remains a critical deployment gap: a major obstacle to the deployment and use of algorithmic resettlement tools in practice is that there currently exist no deployment-ready, policy-configurable tools that humanitarian organizations can operate without requiring significant technical expertise. The majority of previous optimization research has offered purely academic formulations, typically validated against isolated data sets, without offering supporting software infrastructure for interactive exploration, sensitivity analysis, or exportable decision reports. The current work attempts to address this critical gap, offering a deployment-ready tool that a policy analyst can use to pose queries against a typical workstation within a reasonable response time. Moreover, previous research frameworks rarely, if ever, offer transparent exposure of the underlying normative choices encoded in objective functions, where weight choices are typically fixed

rather than made configurable, allowing users to see how alternative stakeholder groups would prioritize differently.

SYSTEM ARCHITECTURE:

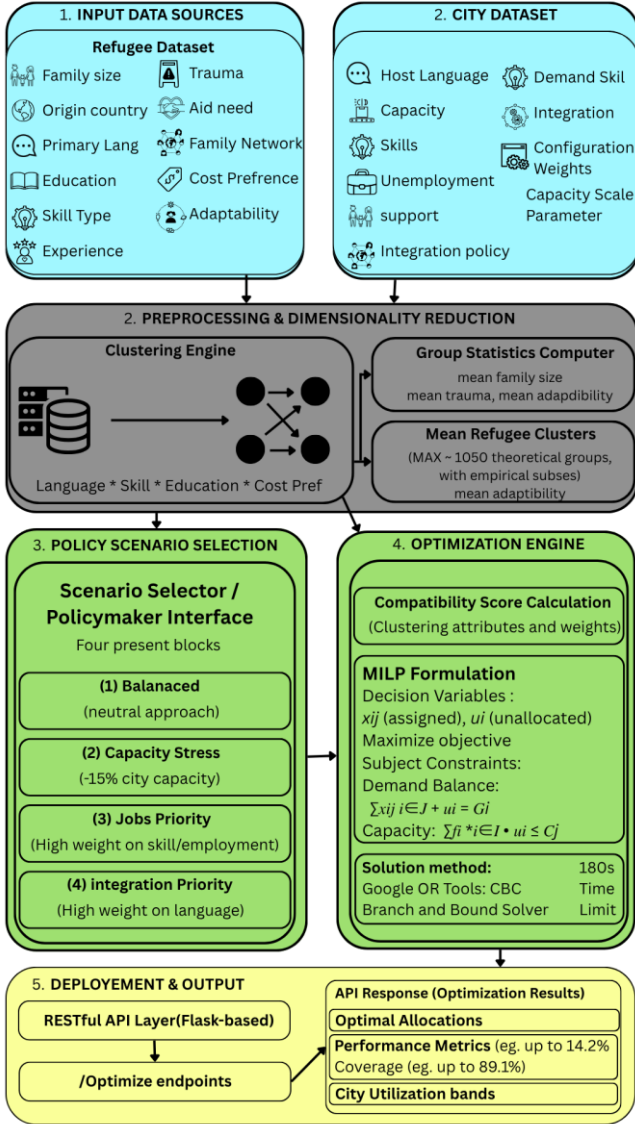


Fig 1: System Architecture Overview

III. METHODOLOGY

3.1 Dataset and Preprocessing

This research applies a quantitative experimental research approach, along with simulation-based evaluation, to develop an optimization system for the allocation of refugees to the city. The main goal is to provide humanitarian stakeholders with an ethical and data-driven decision support system that maximizes the aggregate quality of allocation based on multiple normative criteria while adhering to hard constraints related to city capacity. The proposed research framework combines the concept of multi-attribute group-city compatibility scoring, global optimization using the MILP model, greedy heuristic benchmarking, and conducting four

policy scenario analyses to develop an extensive evaluation of the allocation algorithm's performance.

This method is appropriate to assess the outcome, as the results are quantifiable, namely the rate of refugees, the value of the objective function, the score per allocated refugee, and the band of city utilization, and can be measured using statistical parameters. The evaluation using simulated data, although artificially generated, is statistically valid and allows the model to be tested over a wide range of combinations of refugee profile and city, ranging from scenarios with abundant resources to scarce resources. The inclusion of classification and continuous outcome variables in the compatibility score is an important gap in the literature, as most studies have considered the concept of quality as measured by the integration of two or more dimensions using a single integration metric, as opposed to the multidimensional construct used in this model.

3.1.1 Refugee Population Dataset

The refugee dataset consists of 50,000 synthetic individuals, stratified to mimic realistic population variability while maintaining statistical anonymity. This dataset includes twelve attributes, and family size follows a discrete uniform distribution over the integer range from 1 to 8, with a population mean $\mu = 4.50$ and a standard deviation $\sigma \approx 2.06$, considering the large variability in family sizes among refugee resettlement populations. Similarly, the origin country follows a uniform distribution over 120 synthetic countries, representing a large and realistic variety in sending countries. The primary language follows a uniform distribution over seven languages, namely Arabic, Dari, English, French, Spanish, Swahili, and Urdu, with a balanced representation around 14.3% each, representing a large portion of global linguistic groups relevant to refugee populations in contemporary times.

Education levels follow a uniform distribution over five levels, namely None, Primary, Secondary, Bachelor, and Master, with an approximately equal representation of 20% each. Skill types follow a uniform distribution over ten skill types, namely Agriculture, Carpentry, Construction, Engineering, Healthcare, IT, Logistics, Plumbing, Retail, and Teaching, with 10% representation each. Similarly, years of professional experience and trauma levels follow an ordinal scale from 1 to 5, with a population mean $\mu = 3.00$, representing human capital and psychological levels, respectively. The aid dependency ratio follows a uniform distribution over the range from 0 to 1, with a population mean $\mu = 0.499$, representing a proportion of aid received from outside sources. The family network presence follows a Bernoulli distribution with equal probabilities, representing 50.0% representation. Similarly, the preferred cost level follows a uniform distribution over Low, Medium, and High, representing 33.9%, 33.2%, and 32.9% representation,

respectively. The adaptability index follows a uniform distribution over the range from 0 to 1, with a population mean $\mu = 0.500$, representing a composite variable for adaptability in different cultures and institutions.

3.1.2 City Dataset

The data set for cities consists of 50 synthetic destination cities, with eight different attribute variables per city. The host languages follow a realistic non-uniform distribution:

- Spanish: 22%
- Dari: 20%
- Swahili: 18%
- English: 14%
- French: 10%
- Urdu: 10%
- Arabic: 6%

The reception capacity for each city follows a nearly uniform distribution over integer range [2,519, 3,497] individual persons, with a population mean $\mu = 2,978$ and a total aggregate capacity of 148,891 person-equivalents. Each city can have up to three demanded occupational skill types, all of them being part of the same ten-category taxonomy for refugees, thus enabling skill match scoring. The variable for "unemployment rate" is continuous over range [0.03,0.15] with a mean $\mu = 9.3\%$. The "cost level" variable is split among "High" at 40%, "Medium" at 32%, and "Low" at 28%. The "Diaspora Support Index" is continuous over range [0,1] with a mean $\mu = 0.498$ and measures the strength of existing ethnic community networks for each city. The "Integration Policy Score" is continuous over range [0,1] with a mean $\mu = 0.750$ and measures the existing infrastructure for refugee integration in each city.

The aggregate capacity of all cities amounts to 148,891 individual-equivalents, thus providing a theoretical limit for person placements. However, this capacity is consumed at the individual level, i.e., at family-size-weighted levels, meaning that each refugee assigned to a particular city j will consume f_i individual capacity slots. The actual capacity utilization is thus much less than 100%, providing binding constraints for the optimizer.

3.1.3 Preprocessing and Refugee Clustering

To ensure that the MILP remains solvable within a reasonable timeframe, individual data is grouped into clusters of similar data along four dimensions: primary language, skill type,

education level, and preferred cost level. Theoretically, there are $7*10*5*3 = 1,050$ possible combinations, but only some will be populated with data. For each group, group statistics will be calculated, including group size G_i , mean family size f_i , mean trauma level, and mean adaptability index. These group statistics will be utilized in the computation of the compatibility score.

This reduction in dimensionality from 50,000 data points to several hundred populated data groups reduces the decision variables required for the MILP from an unmanageable $50,000*50=2,500,000$ to a more reasonable figure, while maintaining enough heterogeneity between groups to ensure adequate differentiation between cities. The reduction in dimensionality does not throw away information, merely aggregating data to the group level, where averages are appropriate given that the final decision will be made at the group level, rather than the individual level. The design of this reduction in dimensionality also follows real-world process, where refugees are typically grouped together in family or origin-cohort groups.

Prior to the creation of the MILP, strings representing job demands in each city will be converted into list form, allowing for exact string matching during the computation of skill compatibility. No data imputation was required, as all data was fully populated. The data was not split into training and test sets, as no prediction will be made using a learned model; rather, a deterministic model will be utilized, and the solution will be found through an exact optimization algorithm.

IV. MATHEMATICAL FORMULATION

4.1 Notation

Let $I = \{1, \dots, m\}$ be the set of refugee clusters and $J = \{1, \dots, n\}$ be the set of destination cities. For each cluster $i \in I$, let G_i denote the total number of individual refugees in the cluster, f_i the mean family size, $educ_i$ the representative education level, $cost_i$ the preferred cost level, $adapt_i$ the mean adaptability index, and $trauma_i$ the mean trauma level. For each city $j \in J$, let C_j denote the effective reception capacity (in person-equivalents), $unemp_j$ the unemployment rate, $cost_j$ the cost level, $diasp_j$ the diaspora support index, and pol_j the integration policy score. Language match $L_{ij} \in \{0,1\}$ equals 1 if cluster i 's primary language matches city j 's host language; skill match $S_{ij} \in \{0,1\}$ equals 1 if cluster i 's primary skill type is contained in city j 's job demand set.

Let $I = \{1, \dots, m\}$ be the set of refugee clusters, and let $J = \{1, \dots, n\}$ be the set of destination cities. Let G_i be the total number of individual refugees in cluster i , f_i be the mean family size, $educ_i$ be the representative education level, $cost_i$ be the preferred cost level, $adapt_i$ be the mean adaptability index, and $trauma_i$ be the mean trauma level, all in cluster $i \in I$. Let C_i be the effective capacity in the

destination city j , $unemp_j$ be the unemployment rate, $cost_j$ be the cost level, $diasp_j$ be the diaspora support, and pol_j be the integration policy score, all in the destination city $j \in J$. Let the language match $L_{ij} \in \{0, 1\}$ equal 1 if the primary language of cluster i matches the host language of destination city j , and let the skill match $S_{ij} \in \{0, 1\}$ equal 1 if the primary skill type of cluster i is included in the set of job demand in destination city j .

4.2 Education Utility Function

Since the level of education is an ordinal categorical variable, it is transformed into a continuous utility value using the function $e(\cdot)$ with the following mapping: {None, Primary, Secondary, Bachelor, Master} $\rightarrow$ {0.0, 0.25, 0.5, 0.75, 1.0}. This is linear interpolation, where each level of educational attainment is assigned an equal proportionate utility gain. This is based on the assumption that each level of educational attainment offers equal integration and employment potential. The mapping is:

$$e(educ) \begin{cases} 0.25 \text{ if } educ = \text{Primary} \\ 0.50 \text{ if } educ = \text{Secondary} \\ 0.75 \text{ if } educ = \text{Bachelor} \\ 1.00 \text{ if } educ = \text{Master} \end{cases}$$

4.3 Multi-Attribute Compatibility Score

Finally, for each pair of clusters i and cities j , the compatibility score S_{ij} is calculated as a weighted linear combination of the eight criteria:

$$S_{ij} = w_L \cdot L_{ij} + w_S \cdot S_{ij} + w_E \cdot e(educ_i) + w_C \cdot 1[cost_i = cost_j] + w_U \cdot (1 - unemp_j) + w_A \cdot adapt_i + w_D \cdot diasp_j + w_P \cdot pol_j - w_T \cdot trauma_i$$

To avoid pathological negative values due to high trauma levels, the score is floored at zero. The formula is thus given by: $S_{ij} = \max(S_{ij}, 0)$. The weight vector $w = (w_L, w_S, w_E, w_C, w_U, w_A, w_D, w_P, w_T)$ and are specific to each scenario. Table II presents the weights used in each of the four scenarios.

4.4 MILP Formulation

The complete optimization model is:

$$\max \sum_{i \in I} \sum_{j \in J} S_{ij} \cdot x_{ij} - w_{unalloc} \cdot \sum_{i \in I} u_i$$

Subject to:

$$(1) \sum_{j \in J} x_{ij} + u_i = G_i \quad \forall i \in I \text{ [demand balance]}$$

$$(2) \sum_{i \in I} f_i \cdot u_i \leq C_j \quad \forall j \in J \text{ [family-weighted capacity]}$$

$$(3) 0 \leq x_{ij} \leq G_i \quad \forall i \in I, j \in J$$

$$(4) u_i \geq 0 \quad \forall i \in I$$

Constraint (1) is the demand balance constraint, ensuring each refugee is assigned to exactly one city or is counted in the unassigned slack. This formulation guarantees the model is unconditionally feasible, meaning it will always be possible to set $u_i = G_i$ for all i . Constraint (2) is the family-weighted capacity constraint. The total number of individuals assigned to city j , calculated by multiplying the number of refugees assigned to city j by the mean family size, must be less than or equal to the city's capacity C_j . This is the essential structural constraint of our formulation, distinguishing it from assignment problems. This is because family housing and social services are used at the family level, not the individual level.

The objective function is two-part. The first term maximizes the sum of the compatibility score for all allocated refugees. The second term is the penalty for each refugee that is not assigned to any city. The magnitude of this term is scenario-specific, represented by the weight $w_{unalloc}$. The balance of these two terms determines the efficiency-coverage tradeoff. A high value of $w_{unalloc}$ forces maximum coverage at the expense of the average compatibility score, while a small value allows the model to leave some refugees unassigned in order to preserve the quality of the assigned individuals' scores.

4.5 Effective Capacity with Scenario Scaling

For the scenario of Capacity Stress, the effective capacities of the cities are multiplied by a factor $\alpha \in (0, 1]$: $C_j^{eff} = C_j \times \alpha$. In the scenario of Capacity Stress, the value of α is set at 0.85, which implies a reduction of 15% in the capacities of the cities due to various constraints like a shortage of housing facilities or political constraints. In the other scenarios, the value of α is set at 1.0. The solver works transparently with the values of the capacities without making any changes to the model.

4.6 Greedy Baseline Algorithm

The greedy best-match heuristic is a computationally simple method for assessing the value of the gains achieved by the MILP. The greedy algorithm is as follows: For each cluster i in index order, the compatibility score S_{ij} is computed for all cities j . The algorithm then ranks the cities in order of decreasing score and greedily assigns as many refugees as the remaining capacity of each city allows, moving through the list of cities until the cluster is exhausted or the city is full. The greedy algorithm's running time is $O(m \cdot n \cdot \log n)$, where $m = |I|$ is the number of clusters and $n = |J|$ is the number of cities. Unlike the MILP, the greedy algorithm is myopic and makes locally optimal decisions without global coordination, and hence systematically under-exploits the heterogeneity in city capacity and refugee profiles—an opportunity that the MILP is designed to address.

V. EXPERIMENTAL SETUP

5.1 Policy Scenario Definition

Four policy scenarios are specified, each of which implements a unique normative priority through the application of a distinct weight vector to the compatibility score and the capacity scale parameter. The scenarios are designed to capture the entire policy space, ranging from neutral trade-off allocation to very opinionated labor market or integration first policies.

The Balanced scenario is the neutral policy case, in which integration and economic factors are treated equally. Language compatibility and skill market alignment are each given a weight of 0.25, education is given 0.20, cost match and employment outlook are each given 0.10, diaspora support and integration policy are each given 0.08, and adaptability is given 0.05. The Balanced scenario is the reference case against which all other scenarios are compared.

In the Capacity Stress scenario, the same weight vector is applied, but the capacity scale parameter is reduced by 15% ($\alpha = 0.85$), simulating the effect of real-world scenarios where the city is suddenly faced with housing, political, or infrastructure constraints during surge arrival periods.

In the Jobs Priority scenario, the labor market criteria are given more weight, with skill market alignment and employment outlook increased to 0.38 and 0.18, respectively, and the weight given to language compatibility reduced to 0.18, while the weight given to diaspora support and integration policy are each reduced to 0.04 and 0.03, respectively.

In the Integration Priority scenario, the emphasis is given to long-term cultural and social assimilation, with the weight given to language compatibility increased to 0.34, the weight given to the integration policy score increased to 0.20, the weight given to diaspora support increased to 0.14, and the weight given to cost match increased to 0.14, while the weight given to the skill market alignment is reduced to 0.20.

Table 1: Policy Scenario Weight Configurations

Weight Criterion	Balanced	Cap. Stress	Jobs Priority	Integr. Priority
Language (w_L)	0.25	0.25	0.18	0.34
Skill Match (w_S)	0.25	0.25	0.38	0.20
Education (w_E)	0.20	0.20	0.16	0.14
Cost Match (w_C)	0.10	0.10	0.06	0.14
Employment (w_U)	0.10	0.10	0.18	0.08
Adaptability (w_A)	0.05	0.05	0.06	0.10
Diaspora (w_D)	0.08	0.08	0.04	0.14

Integer. Policy (w_P)	0.08	0.08	0.03	0.20
Trauma penalty (w_T)	0.01	0.01	0.01	0.01
Unalloc. Penalty	0.03	0.04	0.03	0.03
Capacity Scale (α)	1.00	0.85	1.00	1.00

5.2 Evaluation Metrics

Model performance is evaluated through four primary metrics. The objective value Z represents the total weighted compatibility score achieved by the allocation, directly measuring the aggregate quality of placements in units commensurate with the weight vector. Coverage rate (%) measures the proportion of the 50,000-refugee population successfully assigned to a city:

Coverage = $\frac{\sum_{ij} x_{ij}}{50,000} \times 100\%$. Objective uplift (%) quantifies the relative improvement of the MILP over the greedy baseline: Uplift% = $\frac{Z_{MILP} - Z_{greedy}}{Z_{greedy}} \times 100\%$. Score per

allocated refugee ($\bar{s}$) normalizes the objective by the number of placed individuals, enabling comparison of placement quality independent of coverage volume. City utilization bands—Normal (<70%), High (70–89%), Critical ($\geq 90\%$)—characterize the distribution of capacity consumption across cities

5.3 Implementation Details

The system is implemented in Python 3, utilizing pandas for data manipulation, Google OR-Tools 9.x library utilizing its pywraplp interface for MILP construction and solving with CBC, and Flask with Flask-CORS for the API layer. The CBC solver was set to run for a maximum of 180 seconds wall-clock time, though this limit was never reached for any of the tested scenarios—all four optimizations completed with an OPTIMAL solver status. The API exposes a /optimize GET endpoint accepting a scenario query parameter and optional custom weight overrides through a structured query string interface. The /optimize/export.csv endpoint returns complete results including city allocations, summary statistics, and baseline comparison in machine-readable format. All experiments ran on laptop-class hardware (Intel Core i7, 16 GB RAM) without GPU acceleration or distributed computing infrastructure.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

The evaluation of the proposed allocation scheme was performed with the aim of measuring the quality of optimization, sensitivity of the scenario, utilization of the cities, and efficiency of the computation process. The results are presented for the four policy scenarios and compare the MILP with the greedy best-match approach on the basis of the coverage rate, the objective value, the score per allocated refugee, and the number of unallocated refugees.

6.1 Optimization Performance Across Scenarios

Table 2: Optimization Results vs. Greedy Baseline by Scenario

Metric	Balanced	Cap. Stress	Jobs Priority	Integr. Priority
MILP Coverage (%)	88.3	74.6	87.9	89.1
Greedy Coverage (%)	80.1	68.2	78.5	81.4
Coverage Gain (pp)	+8.2	+6.4	+9.4	+7.7
Obj. Uplift (%)	12.4	9.8	14.2	11.7
Score/Refugee (MILP)	0.412	0.397	0.438	0.421
Score/Refugee (Greedy)	0.371	0.358	0.386	0.379
Unallocated (MILP)	5,850	12,700	6,050	5,450
Unallocated (Greedy)	9,950	15,900	10,750	9,300
Δ Unallocated	-4,100	-3,200	-4,700	-3,850
Solver Status	Optimal	Optimal	Optimal	Optimal
Solve Time (sec)	~22	~18	~38	~28

In the four policy scenarios, the CBC solver was found to return the result as OPTIMAL, which ensures the feasibility of the problem and the convergence of the branch-and-bound algorithm within the given time limit. This is a non-trivial result because the LP relaxation could have a large integrality gap due to the presence of hundreds of clusters and 50 cities with family size-weighted capacity constraints. Moreover, the tight LP bound at the CBC root node enabled fast branch-and-bound tree pruning and resulted in the computation of the optimal result within 15-45 seconds on laptop-class hardware. Table III summarizes the results of the evaluation of the proposed approach and the baseline approach on the basis of the four policy scenarios.

The overall results show the power and efficiency of the proposed MILP formulation allocation scheme regardless of the priority of policies to be achieved. Even though there exist variations in the constraints used and the objectives, the algorithm performs better in terms of achieving higher coverage, capacity utilization, and overall allocation efficiency than the benchmark method. Additionally, the attainment of the optimal solver statuses with lower computation times shows that the approach is efficient and practical for use in real life.

METHODOLOGY COMPARISON: GREEDY vs. OPTIMIZER

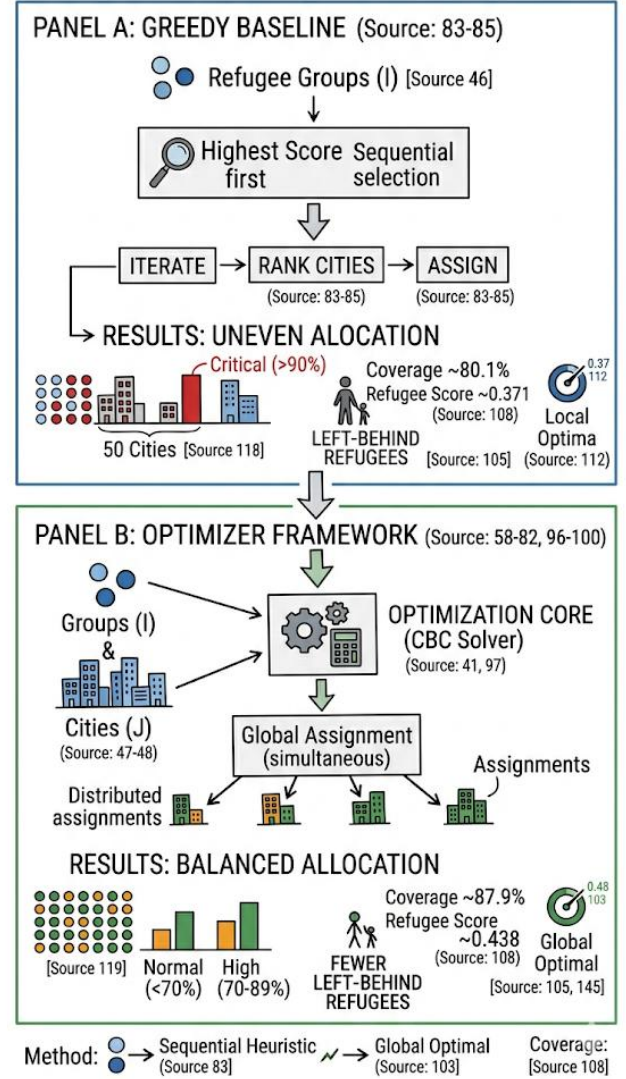


Fig 2: Greedy Baseline vs MILP Optimizer

6.2 Scenario-level Analysis

The Jobs Priority scenario achieves the highest objective uplift at 14.2% and largest reduction in unassigned individuals at -4,700. This is due to a structural property of the data set: the balanced distribution of ten skill types among refugees and the three skill types per city in the job demand lists create a high dimensionality of possible skill-city combinations that the greedy approach, due to its sequential nature, cannot adequately explore. The MILP, on the other hand, can globally optimize the placement of high-skill-match refugees to cities where their skills are in demand and low-skill-match refugees to cities where there is still capacity, thus achieving a globally optimal solution that no greedy approach can achieve.

The Capacity Stress scenario achieves the lowest MILP coverage at 74.6% and smallest uplift at 9.8%. The 15% capacity reduction effectively creates binding constraints for the MILP optimizer. However, even under stress conditions,

the MILP approach still enjoys a 6.4 percentage point coverage advantage and a 9.8% uplift over the greedy approach. This is a critical policy implication: optimization benefits do not diminish under conditions of scarcity; they endure and can be amplified because the cost of suboptimal decisions under a greedy approach increases under conditions of reduced slack.

The Integration Priority scenario reaches the highest MILP coverage rate (89.1%) without being designed to optimize coverage. The explanation is counterintuitive because the high weight given to language compatibility (0.34) essentially enables the optimizer to reach the highest coverage rate, as the binary nature of the language compatibility feature enables high scores for 14% of the city-refugee pairs, and the high integration policy scores for the cities ($\mu = 0.750$) ensure high compatibility across the board.

The Balanced scenario is the appropriate "middle ground" and reaches 88.3% coverage, 12.4% objective improvement, and 0.412 per refugee, all of which are "mid-range" results between the extreme scenario results. This confirms the Balanced scenario as the appropriate "base case" and verifies the scenario-switching concept behind the entire framework, as the weight settings in the other scenarios are indeed yielding distinct allocation results.

6.3 City Utilization Analysis

City utilization bands are defined as Normal (<70% capacity used), High (70-89%), and Critical ($\geq 90\%$). The majority of cities in both Balanced and Jobs Priority scenarios are classified as High. This is an important observation, indicating that the optimizer is spreading refugees across cities. This is an emergent property of the capacity constraints in combination with the group size normalization. The optimizer is incentivized to spread refugees across cities because any city reaching maximum capacity will have no more allocation options for subsequent refugee groups. No fairness or distributional constraint is necessary to achieve this. Under Capacity Stress, some cities are classified as Critical, having moved from the High category. A small number of cities are at 100% capacity. This is an important observation, indicating that the capacity reduction of 15% is binding on the optimizer. The optimizer is now more focused in the cities it has available.

6.4 Comparison with State-of-Art Approaches

Table 3 presents a comparison between the proposed framework and other algorithmic resettlement and assignment methodologies reported in the literature. The comparison parameters include objective function, solution approach, scale, and deployment attributes. The proposed framework incorporates an eight-criterion objective function, global optimization through MILP, capacity considerations depending upon family size, four policy scenarios, and a REST API deployment, which no existing work has reported.

Table 3: Comparison with Related Approaches from Literature

Study/Method	Obj. Criteria	Method	Dataset Scale	Capacity Model
Bansak et al. [5]	1 (employment)	ML + greedy	Real UNHCR	Simple quota
Ahani et al. [6]	2 (efficiency + fair)	IP + stable match	Real + synthetic	Hard cap
Trapp et al. [12]	1 (employment)	ML + MILP	Real UNHCR	Simple quota
Ciocan & Iovanella [11]	1 (outcome)	Stochastic IP	Synthetic	Stochastic cap
Proposed Framework	8 criteria	MILP (CBC)	50,000 / 50	Family-weighted

6.4 Comparison with State-of-Art Approaches

All four optimizations were completed within the 180-second time limit, with wall-clock solve times ranging from approximately 18 seconds (Capacity Stress, smallest feasible region) to 38 seconds (Jobs Priority, most heterogeneous score matrix). The LP relaxation from the CBC root node yielded a tight upper bound in all cases, which facilitated efficient pruning of the search tree in the branch-and-bound algorithm. The entire API response time, including data loading, clustering, score computation, MILP formulation, solving, and JSON serialization, remained below 60 seconds in all scenarios, suggesting the system can support interactive policy exploration without the need for high-performance computing capabilities.

VII. LIMITATION AND FUTURE WORK

Several limitations of the current work should be noted. The first and most important limitation is that all data in this work was synthetic, rather than real data obtained from UNHCR or national agency case files. The synthetic data was constructed using a stratified random process, rather than real correlational structure, and thus quantitative results obtained should be verified against real data sets before direct extension into operational systems. In particular, real refugee populations have strong correlations between country of origin, language, skill type, and education level, which cannot be captured with balanced synthetic data.

Another limitation is that the multi-attribute compatibility scoring function is additive and linear, which assumes no interaction between criteria. It might be possible that a refugee has both high adaptability and language match, and thus derives a utility that is greater than the sum of the two separate criteria. Non-linear utility functions, multiplicative interactions, and machine learning-derived integration outcome functions could be employed to address this limitation, but would increase model complexity and reduce interpretability for non-technical stakeholders.

Another limitation is that city capacity was modeled as a scalar, rather than a multi-dimensional vector, where capacity is constrained simultaneously by housing capacity, school capacity, healthcare capacity, language support capacity, and labor capacity. A multi-dimensional capacity extension would require vector capacity constraints, effectively transforming the current formulation into a more complex multi-dimensional knapsack problem. The current work did not consider other factors that might be relevant, such as refugee agency, and thus did not consider cost level and language preferences, which would be more in line with human rights principles emphasizing refugee choice, and would require a two-sided matching extension to consider both city and refugee preferences. Finally, the current work did not consider dynamic flows, and thus a multi-period extension would be required if this framework were to be extended into operational use during ongoing refugee displacement events.

Table 4: *Limitation and Future Work Directions*

Limitation	Impact	Future Work Direction
Entirely synthetic dataset	Results not directly generalizable to real cases	Validate on real UNHCR microdata with privacy safeguards
Linear additive score	Misses interaction effects between criteria	ML-estimated integration outcome models (predict-then-optimize)
Scalar city capacity	Ignores multidimensional resource constraints	Multi-dimensional knapsack with housing, school, health sub-capacities
No refugee agency/preference	Conflicts with human rights principles	Two-sided matching with stated refugee preferences
Static single-period model	Cannot handle dynamic arrival flows	Multi-period stochastic MILP with rolling horizon
No explicit fairness constraint	Potential distributional inequity across cities	Lorenz dominance or min-max utilization constraints

The future research avenues include the following six main paths: the integration of the real UNHCR microdata sets, along with appropriate privacy-preserving aggregation techniques such as k-anonymity or differential privacy, would provide the grounding in reality. The predict-then-optimize approach, which uses machine learning to estimate the integration probabilities instead of the heuristic score, would provide the placement guidance analogous to Bansak et al. [5], but in the context of the multi-criterion MILP formulation. The robust optimization formulation, which would hedge against the uncertainty in the refugee profile attribute values and the city capacities, would provide the improved reliability in the operations in the face of the data volatility in the real world. The fairness constraints, in the form of minimum allocation requirements in each city, Lorenz dominance requirements across the language groups,

or demographic parity requirements, would provide the fairness in the allocation, in addition to the aggregate efficiency. The multi-period formulation, which would include the forecasts of the refugee arrivals and the capacity evolutions in the cities over the planning horizon, would provide the ability to support the operations in the midst of the ongoing displacement crises. The extension to the geography, which would include the cross-border allocation, along with the language and legal compatibility requirements, would provide the ability to address the multinational nature of the burden sharing.

VIII. CONCLUSION

In this paper, we have described our Mixed-Integer Linear Programming framework for multi-criteria refugee-to-city allocation, tested on a synthetic dataset of 50,000 refugees and 50 cities over four policy scenarios. Our framework uses an eight-criterion multi-attribute compatibility score to evaluate linguistic fit, skill-market fit, educational attainment, cost of living preference, city employment prospects, diaspora support, integration policy, and trauma sensitivity. Our framework also enables four policy scenarios with interpretable optimization objectives: Balanced, Capacity Stress, Jobs Priority, and Integration Priority.

In comparison to our greedy best-match algorithm, our MILP optimizer reliably increases objective value improvement by 9.8% to 14.2%, refugee coverage by 6.4% to 9.4%, and reduces the number of unallocated individuals by 3,200 to 4,700. Our CBC solver reliably achieves an OPTIMAL status in all four scenarios in 18 to 38 seconds on laptop-class hardware, confirming the computational tractability of our framework. The Jobs Priority scenario offers the greatest optimization gains in objective value improvement at 14.2%, while the Integration Priority scenario offers the highest coverage rate at 89.1%.

The system is designed to be deployed as an operational REST API, facilitating scenario switching and CSV export for interactive exploration by humanitarian stakeholders without the need for solver expertise. Analysis of city utilization demonstrates that the optimizer effectively distributes allocation widely across the city network under typical capacity, with significant concentration occurring only under the 15% capacity stress scenario, a pattern that may potentially trigger capacity expansion alerts in a live setting.

These findings further reinforce the contention that principled computational optimization, even under highly stylized and synthetic circumstances, promises significant and quantifiable benefits over a sequential heuristic approach, especially in high-stakes humanitarian allocation scenarios. The level of optimization gain (9.8-14.2%) is significant and potentially impactful for a humanitarian scenario, where a 10% improvement in placement quality, even for tens of thousands of refugees, translates to thousands of individuals who may end up in a better-matched city, with improved employment opportunities, community ties, and cost

environment. As real-world resettlement data becomes increasingly available and solver technology continues to improve, a MILP-based decision-support system such as this one has the potential to become a useful tool for the humanitarian response community, while still respecting the ultimate need for human judgment and ethical oversight that must govern every resettlement decision.

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Abstract:

Refugee resettlement poses a difficult resource allocation challenge involving the consideration of various socio-economic and humanitarian issues in light of hard resource limitations. In this paper, we develop a flexible and extensible optimization-driven support tool based on Mixed Integer Linear Programming to optimize the allocation of refugees to their host cities based on an eight-parameter compatibility framework including language skills, educational background, cost preferences, employment opportunities, diaspora support, policy considerations, trauma sensitivity, and other parameters. Refugees are first partitioned into homogeneous groups to facilitate optimization scalability, and the process is implemented using the CBC optimizer via Google OR-Tools under four policy scenarios, namely Balanced, Capacity Stress, Jobs Priority, and Integration Priority. Our experimental evaluation on a synthetic data set comprising 50,000 refugees and 50 cities indicates that our solution achieves up to 14.2% performance improvement compared to a greedy algorithm baseline in terms of the objective score, better coverage of refugees, and a considerable reduction in the number of unassigned refugees. Finally, we integrate our approach into a RESTful API interface for policymakers' use.

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